Financial Econometrics Practice TS #2 (Forecasting)

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1 Load Packages and Data

```r
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.5.2
```

```r
library(tseries)
```

```
## Warning: package 'tseries' was built under R version 4.5.2

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
library(lmtest)
```

```
## Loading required package: zoo

## Warning: package 'zoo' was built under R version 4.5.2

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(nortest)
library(MASS)
```

Read the data as a time series object. The data is the number of individuals registered as unemployed in SEPE offices since January 2000 available at https://expinterweb.mites.gob.es/series/. Select window starting 2000.

```r
series <- window(
  ts(read.table("unemploy.dat", dec = ",") / 1e6,
     start = 1996, freq = 12),
  start = c(2000),
  end = c(2025, 12)
)

# Display properties
cat("Class:", class(series), "\n")
```

```
## Class: ts
```

```r
cat("Length:", length(series), "observations\n")
```

```
## Length: 312 observations
```

```r
cat("Start:", start(series)[1], "Period", start(series)[2], "\n")
```

```
## Start: 2000 Period 1
```

```r
cat("End:", end(series)[1], "Period", end(series)[2], "\n")
```

```
## End: 2025 Period 12
```

```r
cat("Frequency:", frequency(series), "\n\n")
```

```
## Frequency: 12
```

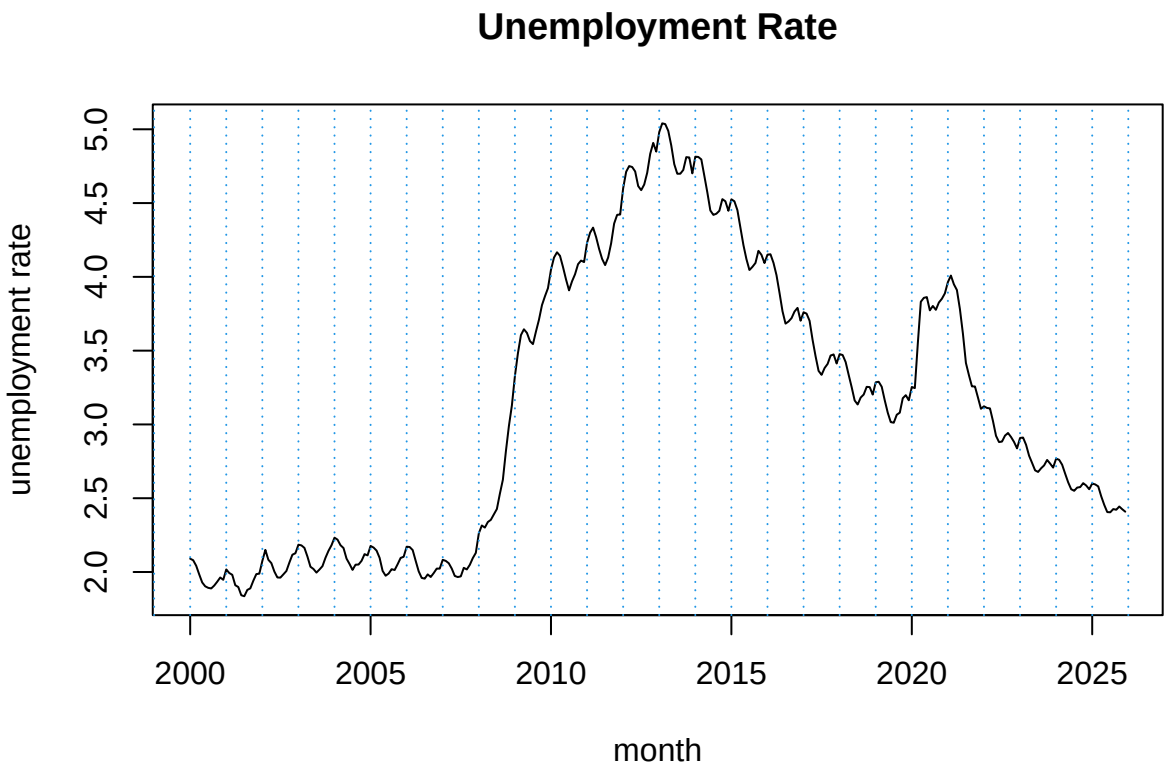
```r
# show values
round(series, 3)
```

```
##        Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct   Nov   Dec
## 2000 2.090 2.080 2.042 1.984 1.929 1.903 1.893 1.888 1.908 1.934 1.963 1.947
## 2001 2.017 1.993 1.981 1.910 1.898 1.843 1.836 1.879 1.889 1.941 1.986 1.989
## 2002 2.075 2.150 2.083 2.060 2.003 1.963 1.962 1.984 2.007 2.065 2.117 2.127
## 2003 2.185 2.180 2.163 2.104 2.036 2.020 1.996 2.017 2.040 2.097 2.143 2.181
## 2004 2.233 2.219 2.182 2.162 2.091 2.054 2.014 2.050 2.051 2.076 2.121 2.113
## 2005 2.177 2.165 2.145 2.096 2.007 1.975 1.989 2.019 2.013 2.053 2.096 2.103
## 2006 2.172 2.169 2.149 2.076 2.005 1.960 1.955 1.984 1.966 1.993 2.023 2.023
## 2007 2.083 2.075 2.059 2.023 1.973 1.966 1.970 2.028 2.017 2.049 2.094 2.130
## 2008 2.262 2.315 2.301 2.339 2.354 2.390 2.427 2.530 2.625 2.818 2.989 3.129
## 2009 3.328 3.482 3.605 3.645 3.620 3.565 3.544 3.629 3.709 3.808 3.869 3.924
## 2010 4.048 4.131 4.167 4.142 4.066 3.982 3.909 3.970 4.018 4.086 4.110 4.100
## 2011 4.231 4.299 4.334 4.269 4.190 4.122 4.080 4.131 4.227 4.361 4.420 4.422
## 2012 4.600 4.712 4.751 4.744 4.714 4.615 4.587 4.626 4.705 4.834 4.908 4.849
## 2013 4.981 5.040 5.035 4.989 4.891 4.764 4.699 4.699 4.724 4.811 4.809 4.701
## 2014 4.814 4.812 4.796 4.684 4.572 4.450 4.420 4.428 4.448 4.527 4.512 4.448
## 2015 4.526 4.512 4.452 4.333 4.215 4.120 4.046 4.068 4.094 4.176 4.149 4.094
## 2016 4.151 4.153 4.095 4.011 3.891 3.767 3.683 3.697 3.720 3.765 3.790 3.703
## 2017 3.760 3.751 3.702 3.573 3.461 3.363 3.336 3.382 3.410 3.467 3.474 3.413
## 2018 3.477 3.470 3.423 3.336 3.252 3.162 3.135 3.182 3.203 3.255 3.253 3.202
## 2019 3.286 3.289 3.255 3.164 3.079 3.016 3.011 3.066 3.080 3.178 3.198 3.164
## 2020 3.254 3.246 3.548 3.831 3.858 3.863 3.773 3.803 3.776 3.826 3.851 3.888
## 2021 3.964 4.009 3.950 3.911 3.781 3.614 3.416 3.334 3.258 3.257 3.183 3.106
## 2022 3.123 3.112 3.109 3.023 2.923 2.881 2.884 2.924 2.942 2.915 2.881 2.838
## 2023 2.908 2.911 2.862 2.788 2.739 2.689 2.678 2.703 2.722 2.759 2.735 2.707
## 2024 2.768 2.760 2.727 2.667 2.608 2.561 2.550 2.572 2.575 2.602 2.586 2.561
## 2025 2.599 2.593 2.580 2.513 2.455 2.406 2.405 2.427 2.422 2.444 2.425 2.409
```

```r
# constants
series_name <- "Unemployment Rate"
start_year <- 2000
start_period <- 1
freq <- 12
period_length <- 12
xlabel <- "month"
ylabel <- "unemployment rate"
```

some obvious yearly seasonality, but SARIMA will handle that

```r
plot(series, main=series_name, xlab=xlabel, ylab=ylabel)
abline(v=1996:2026,col=4,lty=3)
```



To model, we first achieve variance constancy and let SARIMA handle the seasonality and mean stability.

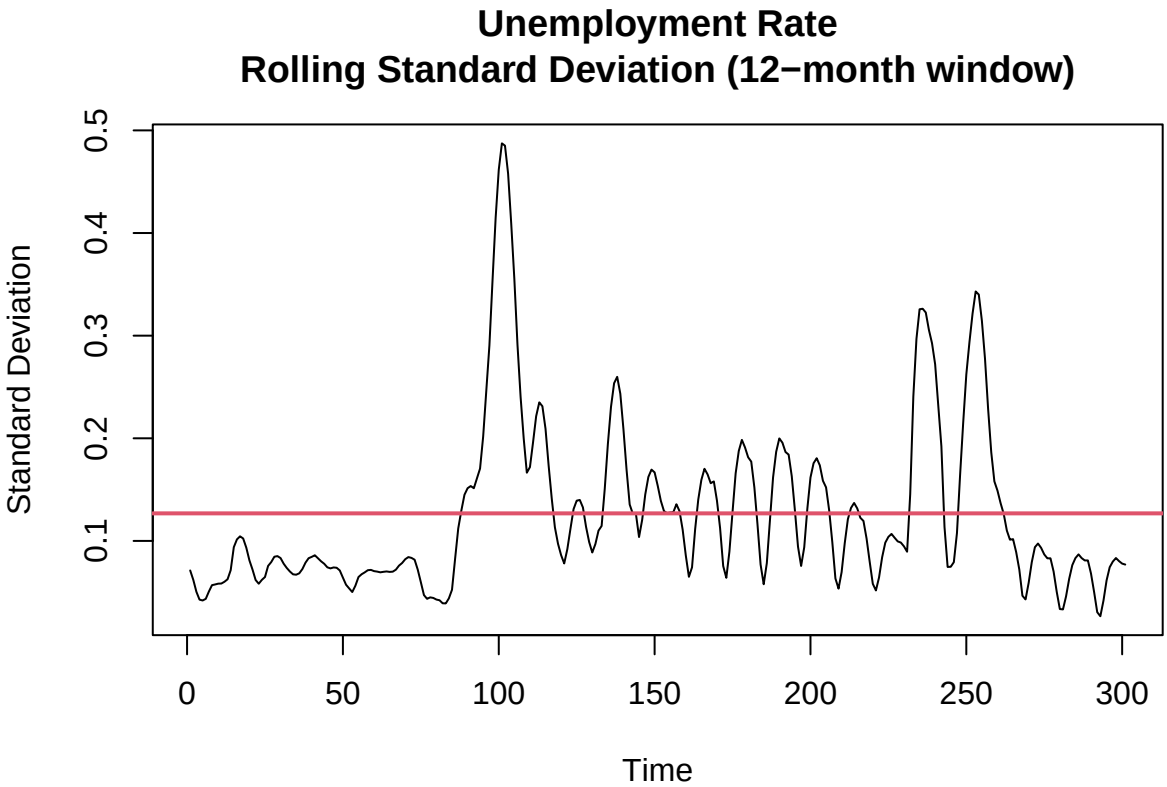
2 Transformation to Variance Constancy

use rolling window stdev better than boxplot, with heteroskedasticity

```
#function to plot rolling standard deviation
plot_rolling_std <- function(series, window_size, name="Series") {
  rolling_std <- sapply(1:(length(series) - window_size + 1), function(i) {
    sd(series[i:(i + window_size - 1)])
  })

  plot(rolling_std, type = "l",
    main = paste(name, "\nRolling Standard Deviation (",
           window_size, "-month window)", sep = ""),
    xlab = "Time", ylab = "Standard Deviation")
  abline(h=mean(rolling_std), col=2, lwd=2)
}
```

2.1 Rolling standard deviation

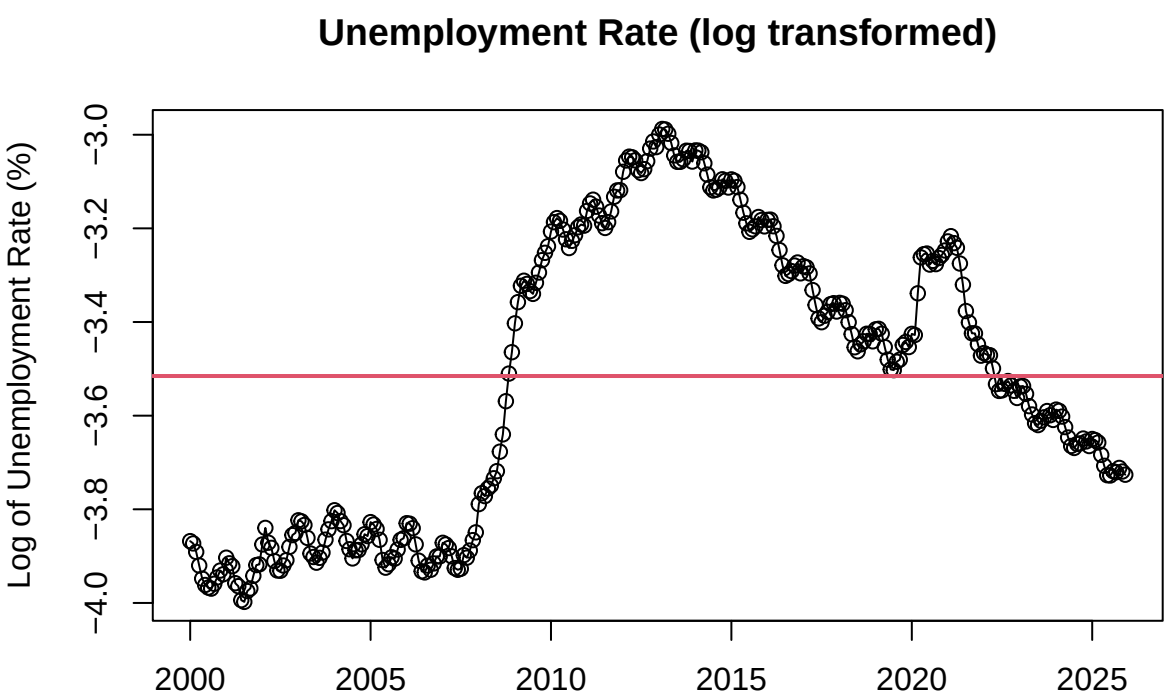
```
plot_rolling_std(series, 12, series_name)
```



##

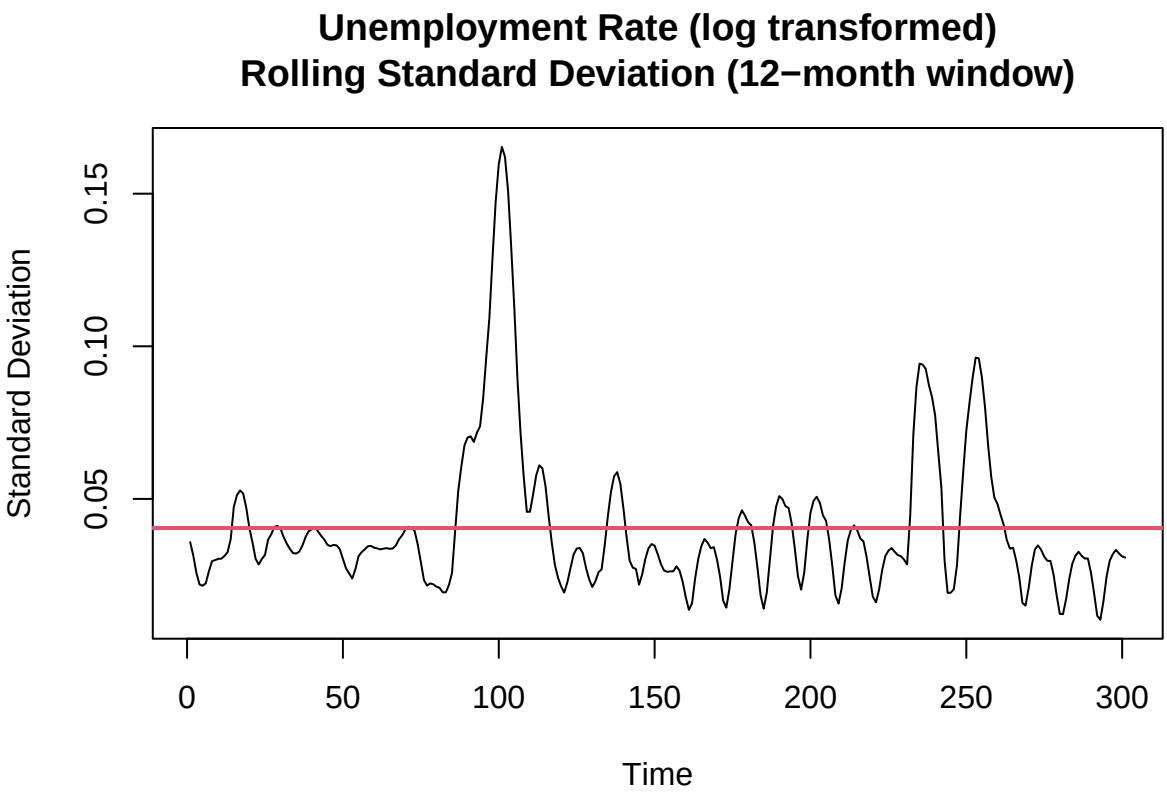
LN Transformation seemed to handle mean stability and seasonality

```
ln_transformed <- log(series / 100)
plot(ln_transformed, type="o",
  main=paste(series_name, "(log transformed)"),
  ylab=paste("Log of Unemployment Rate (%)", sep=""),
  xlab="")
abline(h=mean(ln_transformed), col=2, lwd=2)
```



still with obvious heteroskedasticity here

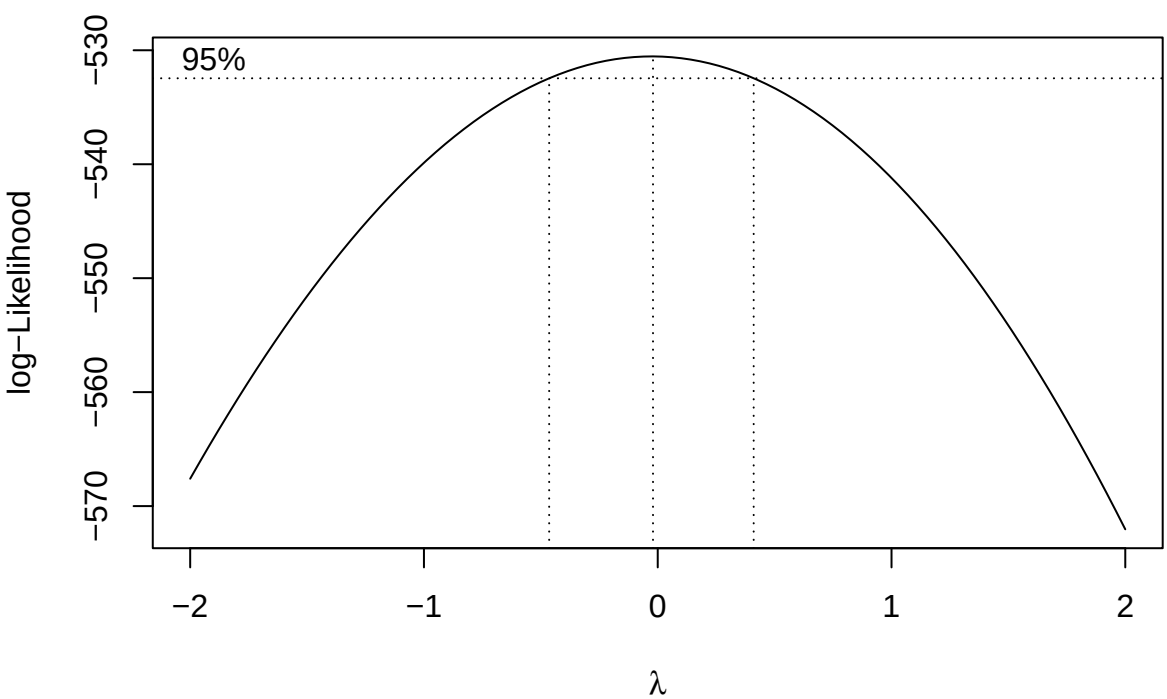
```
plot_rolling_std(ln_transformed, 12, paste(series_name, "(log transformed)"))
```



##

Box Cox Transformation

```
# Box-Cox transformation to find optimal lambda for variance stabilization
boxcox_result <- boxcox(series ~ 1, lambda = seq(-2, 2, by = 0.1))
```

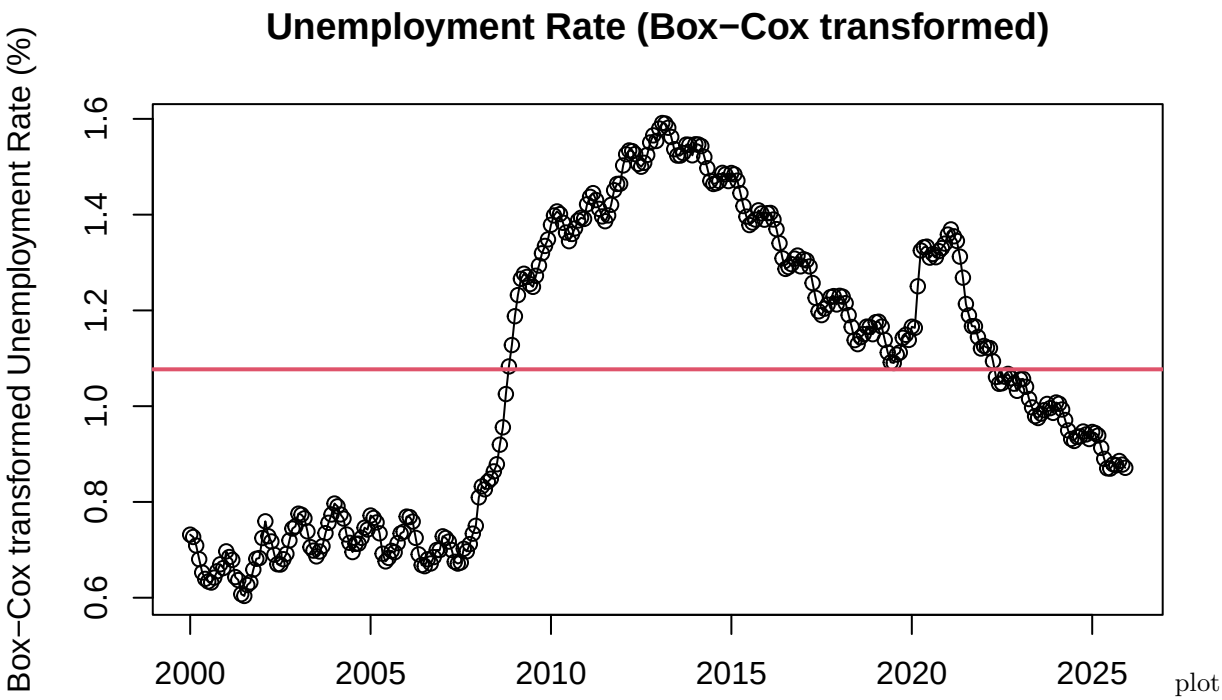


```r
optimal_lambda <- boxcox_result$x[which.max(boxcox_result$y)]
cat("Optimal lambda for Box-Cox transformation:", optimal_lambda, "\n")
```

```
## Optimal lambda for Box-Cox transformation: -0.02020202
```

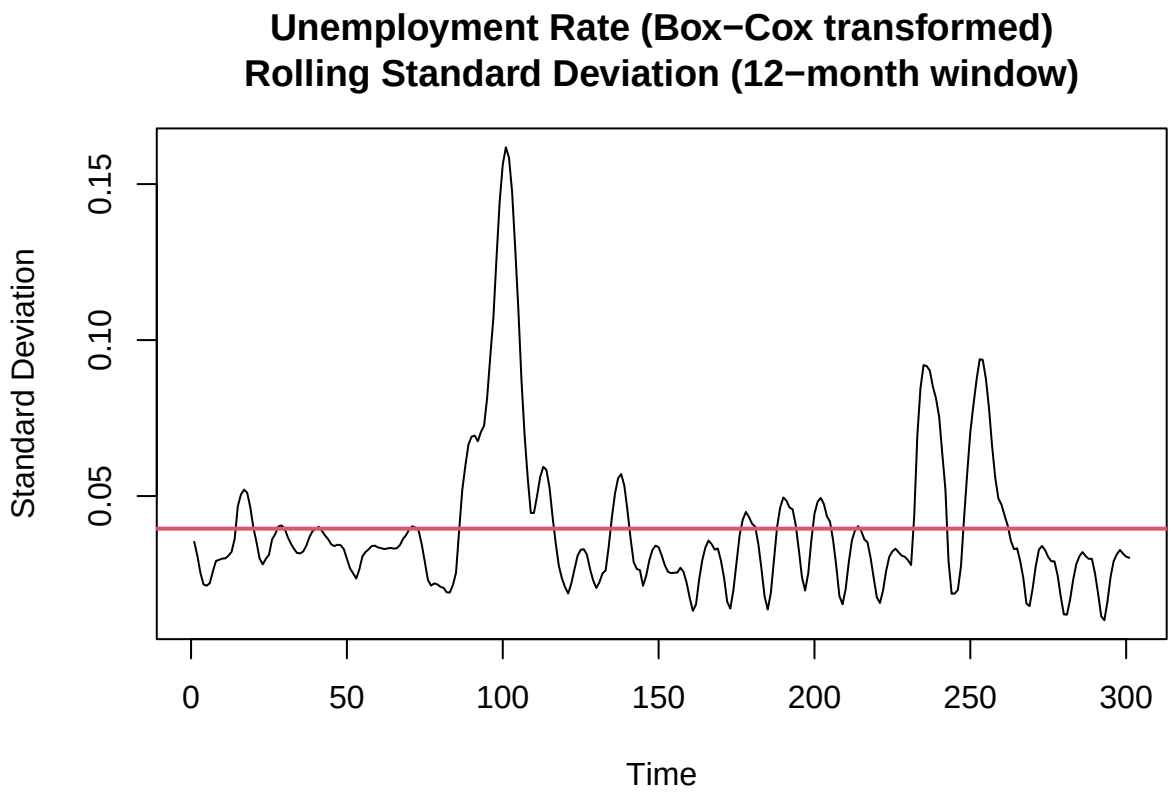
```r
# Apply Box-Cox transformation with optimal lambda
bc_transformed <- (series^optimal_lambda - 1) / optimal_lambda
```

```r
plot(bc_transformed, type="o",
  main=paste(series_name, "(Box-Cox transformed)"),
  ylab=paste("Box-Cox transformed Unemployment Rate (%)", sep=""),
  xlab="")
abline(h=mean(bc_transformed), col=2, lwd=2)
```



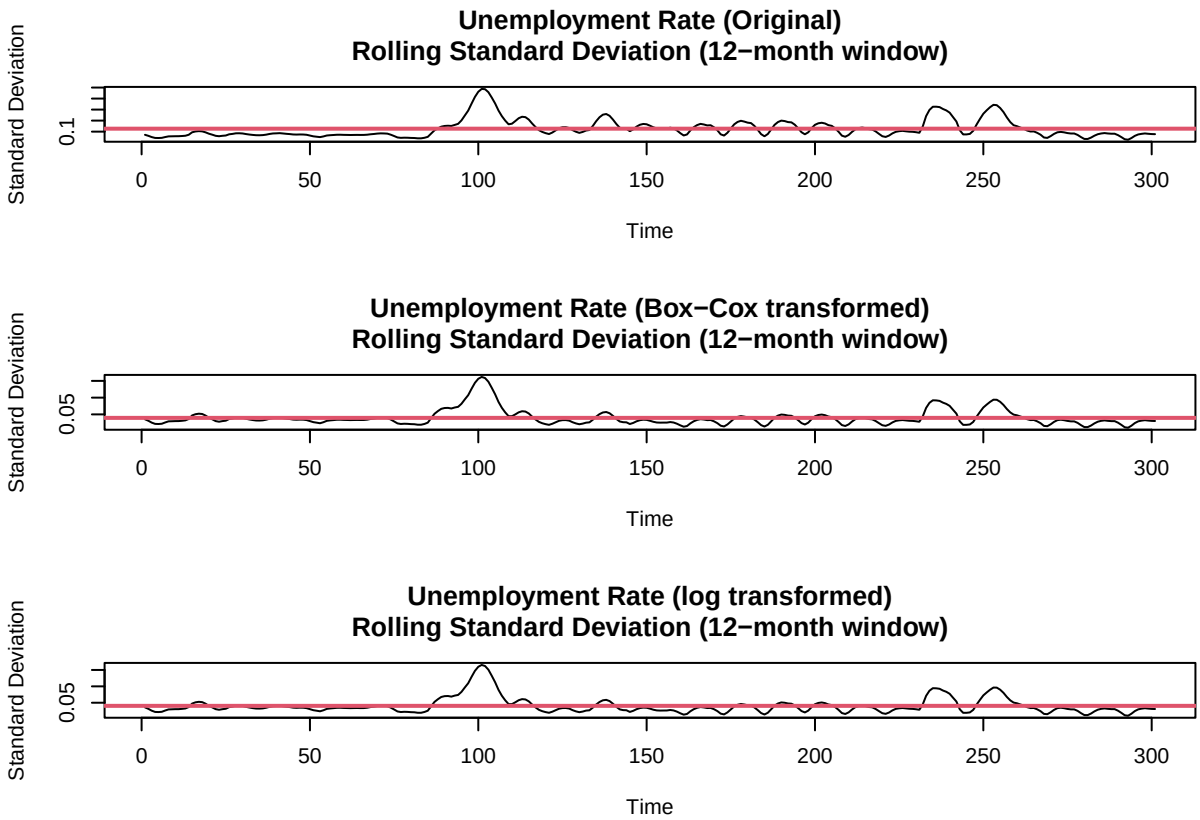
plot

```r
plot_rolling_std(bc_transformed, 12, paste(series_name, "(Box-Cox transformed)"))
```



plot all three transformations together for comparison

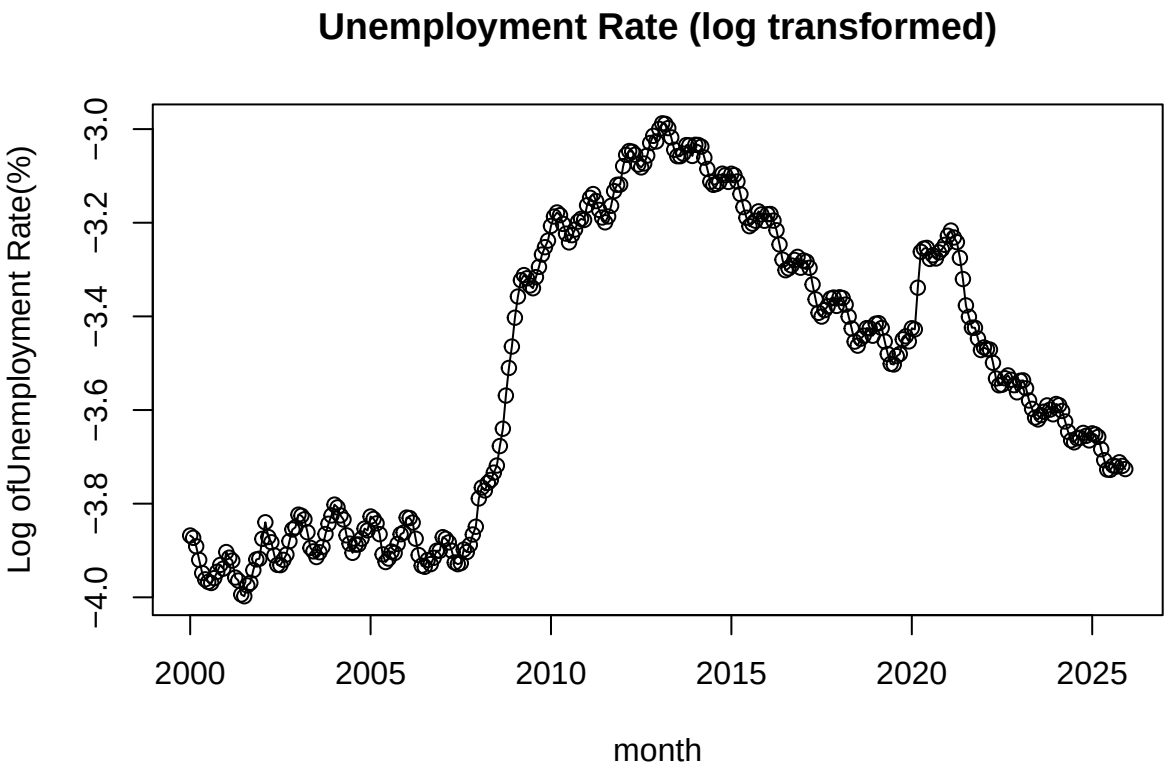
```
par(mfrow=c(3,1))
plot_rolling_std(series, 12, paste(series_name, "(Original)"))
plot_rolling_std(bc_transformed, 12, paste(series_name, "(Box-Cox transformed)"))
plot_rolling_std(ln_transformed, 12, paste(series_name, "(log transformed)"))
```



#

Prep for SARIMA

```
par(mfrow=c(1,1))
working_series <- ln_transformed
working_series_name <- paste(series_name, "(log transformed)")
ylabel <- paste("Log of", series_name, "(%)", sep="")
plot(working_series, type="o", main=working_series_name, ylab=ylabel, xlab=xlabel)
```



ACF

PACF to see potential sarima models

definitely need differencing

```
par(mfrow=c(1,2))
acf(working_series, ylim=c(-1,1), lag.max=max(60, 3*freq),
    lwd=2, main=paste("ACF:", working_series_name),
)
pacf(working_series, ylim=c(-1,1), lag.max=max(60, 3*freq),
     lwd=2, main=paste("PACF:", working_series_name),
     col=c(2, rep(1, 59)))
```

```r
par(mfrow=c(1,1))
```

3 SARIMA Modeling

3.1 SARIMA function

```r
analyze_sarima <- function(series, order, seasonal) {

  # --- Setup ---
  s <- frequency(series)
  n <- length(series)
  model_name <- paste0("ARIMA(", paste(order, collapse=","), ")(",
                       paste(seasonal$order, collapse=","), ")", s)

  cat("\n==================================================================\n")
  cat("  ANALYSIS FOR MODEL:", model_name, "\n")
  cat("==================================================================\n")

  # ----------------------------------------------------------------------
  # 1. Estimation & Significance Check
  # ----------------------------------------------------------------------
  fit_full <- arima(series, order = order, seasonal = seasonal)

  diag_var <- diag(fit_full$var.coef)
  if (any(diag_var <= 0)) {
    cat("\n  CRITICAL WARNING: The Hessian matrix is unstable (Negative Variances).\n")
    se_vec <- ifelse(diag_var > 0, sqrt(diag_var), NA)
  } else {
```

```r
  se_vec <- sqrt(diag_var)
}

t_vec  <- fit_full$coef / se_vec
p_vec  <- 2 * (1 - pnorm(abs(t_vec)))

coeftest <- data.frame(
  Estimate = fit_full$coef,
  Std.Error = se_vec,
  t_stat = t_vec,
  p_value = p_vec
)

coeftest$Significant <- ifelse(!is.na(coeftest$p_value) &
                                 (coeftest$p_value < 0.05 | abs(coeftest$t_stat) > 2),
                               "Yes", "No")

cat("\n[1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):\n")
print(round(coeftest[,1:4], 4))
cat("\nSignificance Flags:\n")
print(coeftest[,5, drop=FALSE])

# ----------------------------------------------------------------------
# 2. Residual Analysis (Diagnostics)
# ----------------------------------------------------------------------
resid <- residuals(fit_full)

# A. Causality / Invertibility Roots (Min Modulus)
# AR Roots
ar_poly <- fit_full$model$phi
if (length(ar_poly) > 0 && any(ar_poly != 0)) {
  ar_roots_mod <- Mod(polyroot(c(1, -ar_poly)))
  min_ar_root  <- min(ar_roots_mod)
} else {
  min_ar_root  <- NA
}

# MA Roots
ma_poly <- fit_full$model$theta
if (length(ma_poly) > 0 && any(ma_poly != 0)) {
  ma_roots_mod <- Mod(polyroot(c(1, ma_poly)))
  min_ma_root  <- min(ma_roots_mod)
} else {
  min_ma_root  <- NA
}

is_causal     <- ifelse(!is.na(min_ar_root), min_ar_root > 1, TRUE)
is_invertible <- ifelse(!is.na(min_ma_root), min_ma_root > 1, TRUE)

cat("\n[2] DIAGNOSTICS:\n")
cat("Min AR Root Modulus: ", min_ar_root, "\n")
cat("Min MA Root Modulus: ", min_ma_root, "\n")
```

```r
# B. Plots
par(mfrow=c(2,2), mar=c(4,4,3,2))
plot(resid, main="Residuals", ylab="Error")
abline(h=0, col="gray"); abline(h=c(-3*sd(resid), 3*sd(resid)), lty=2, col="red")
scatter.smooth(sqrt(abs(resid)), main="Variance Check", ylab="sqrt(|Resid|)", lpars=list(col="red"))
qqnorm(resid); qqline(resid, col="red")
hist(resid, breaks=20, freq=FALSE, main="Histogram"); curve(dnorm(x, mean=mean(resid), sd=sd(resid)),

# C. Tests
sw_test <- shapiro.test(resid)
lb_test <- Box.test(resid, lag = 20, type = "Ljung-Box")

# ----------------------------------------------------------------------
# 3. Forecasting & Cross-Validation
# ----------------------------------------------------------------------
train_end <- c(floor(time(series)[n-12]), cycle(series)[n-12])
train_set <- window(series, end = train_end)
test_set  <- window(series, start = time(series)[n-11])

fit_train <- arima(train_set, order = order, seasonal = seasonal)
fc <- predict(fit_train, n.ahead = 12)

fc_orig <- exp(fc$pred); test_orig <- exp(test_set)
lower_bound <- exp(fc$pred - 1.96 * fc$se); upper_bound <- exp(fc$pred + 1.96 * fc$se)

# ----------------------------------------------------------------------
# 4. Summary Table Generation
# ----------------------------------------------------------------------
summary_row <- data.frame(
  Model_Name      = model_name,
  AIC             = fit_full$aic,
  BIC             = BIC(fit_full),
  Sigma2          = fit_full$sigma2,

  Resid_Norm_PVal = sw_test$p.value,
  Normal       = ifelse(sw_test$p.value > 0.05, "Yes", "No"),

  Resid_Ind_PVal  = lb_test$p.value,
  Independent  = ifelse(lb_test$p.value > 0.05, "Yes", "No"),

  # Positioned as requested: Root value to the left of the Boolean status
  Min_AR_Root     = min_ar_root,
  Causal          = ifelse(is_causal, "Yes", "No"),

  Min_MA_Root     = min_ma_root,
  Invertible      = ifelse(is_invertible, "Yes", "No"),

  RMSE            = sqrt(mean((test_orig - fc_orig)^2)),
  MAE             = mean(abs(test_orig - fc_orig)),
  MAPE_Pct        = mean(abs((test_orig - fc_orig) / test_orig)) * 100,
  Mean_CI_Width   = mean(upper_bound - lower_bound)
)
```

```r
  cat("\n[3] PERFORMANCE SUMMARY:\n")
  print(t(summary_row))

  return(summary_row)
}
```

3.2 Comparison Function

```r
# -- COMPARATIVE ANALYSIS FUNCTION --


compare_sarima_models <- function(series, models_list) {

  cat("\n--- Starting Automated Model Comparison ---\n")

  # This will store the summary row for each successful model
  comparison_results <- list()

  # Loop through each model specification in the list
  for (i in seq_along(models_list)) {
    spec <- models_list[[i]]

    cat(paste0("\nProcessing Model ", i, " of ", length(models_list), "...\n"))

    # Use tryCatch to prevent the loop from stopping if a model fails to converge
    result <- tryCatch({
      analyze_sarima(series = series,
                     order = spec$order,
                     seasonal = spec$seasonal)
    }, error = function(e) {
      cat(paste0("  Error in Model ", i, ": ", e$message, "\n"))
      return(NULL) # Return NULL so we can skip it
    })

    # If the model was successful, add it to our list
    if (!is.null(result)) {
      comparison_results[[length(comparison_results) + 1]] <- result
    }
  }

  # Combine all the rows into one master table
  final_table <- do.call(rbind, comparison_results)

  # Sort by BIC by default (it penalizes complexity more than AIC)
  final_table <- final_table[order(final_table$BIC), ]

  cat("\n--- Comparison Complete ---\n")
  return(final_table)
}
```

3.3 Models

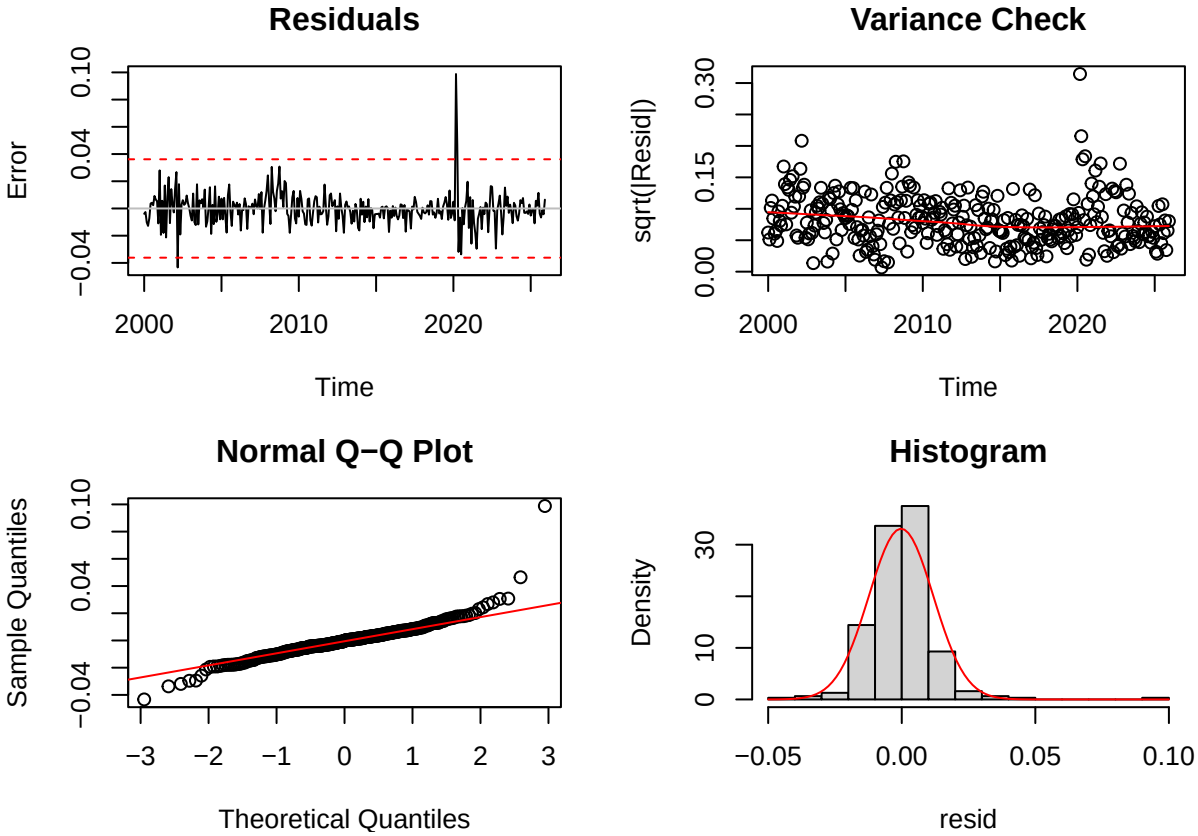
```r
# Specify Models (for now I just made up some)
my_models <- list(
  list(order = c(1,1,1),
       seasonal = list(order = c(2,0,2), period = 12)),

  list(order = c(1,1,2),
       seasonal = list(order = c(2,0,2), period = 12)),

  list(order = c(0,1,9),
       seasonal = list(order = c(2,0,2), period = 12)),

  list(order = c(7,1,0),
       seasonal = list(order = c(2,0,2), period = 12))
)
```

3.4 Run all models

```r
# Run Analysis
comparison_table <- compare_sarima_models(working_series, my_models)
```

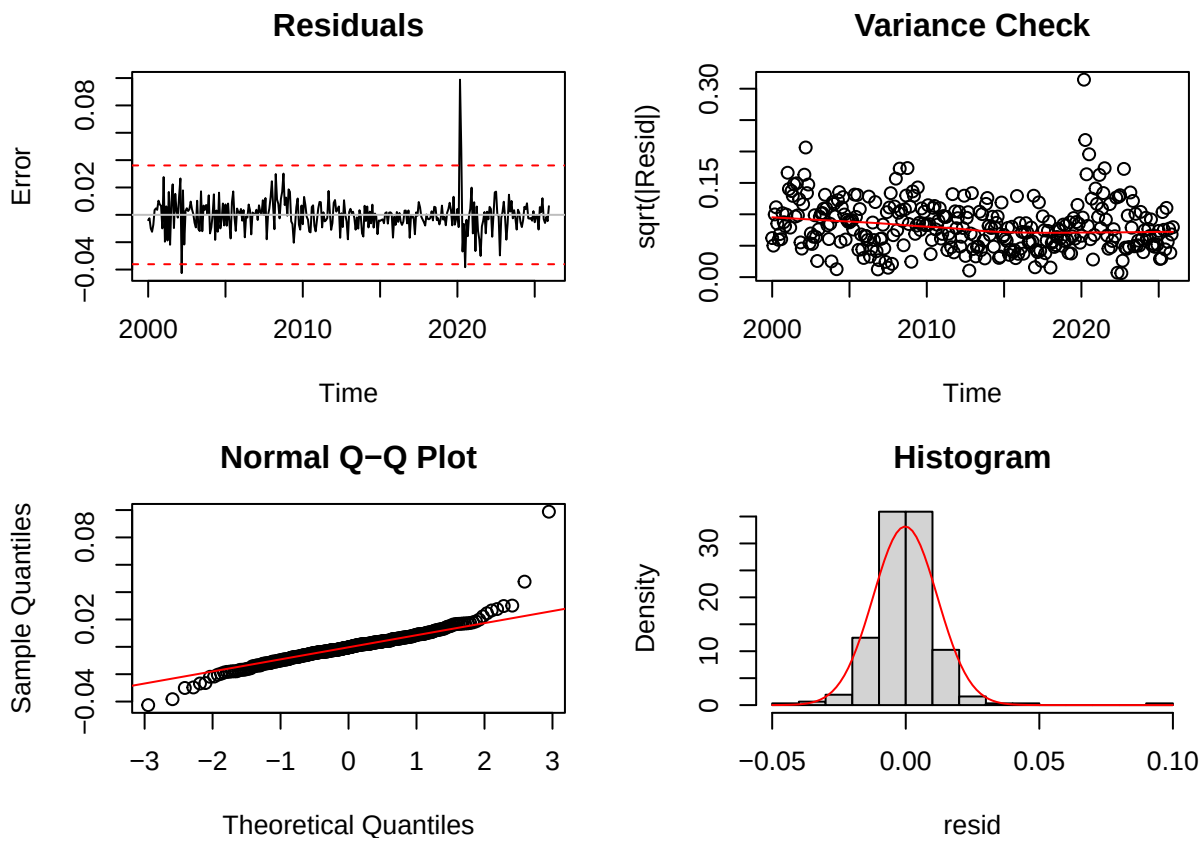
```
##
## --- Starting Automated Model Comparison ---
##
## Processing Model 1 of 4...
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(1,1,1)(2,0,2)12
## ==================================================================
##
##   CRITICAL WARNING: The Hessian matrix is unstable (Negative Variances).
## Warning in sqrt(diag_var): NaNs produced
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##      Estimate Std.Error  t_stat p_value
## ar1    0.7462    0.0634 11.7626   0.000
## ma1   -0.2191    0.0978 -2.2406   0.025
## sar1   0.2347        NA      NA      NA
## sar2   0.7504        NA      NA      NA
## sma1  -0.0927        NA      NA      NA
## sma2  -0.6525        NA      NA      NA
##
## Significance Flags:
##      Significant
## ar1          Yes
## ma1          Yes
## sar1          No
## sar2          No
## sma1          No
## sma2          No
##
## [2] DIAGNOSTICS:
```

```
## Min AR Root Modulus:  1.000709
## Min MA Root Modulus:  1.013097
```



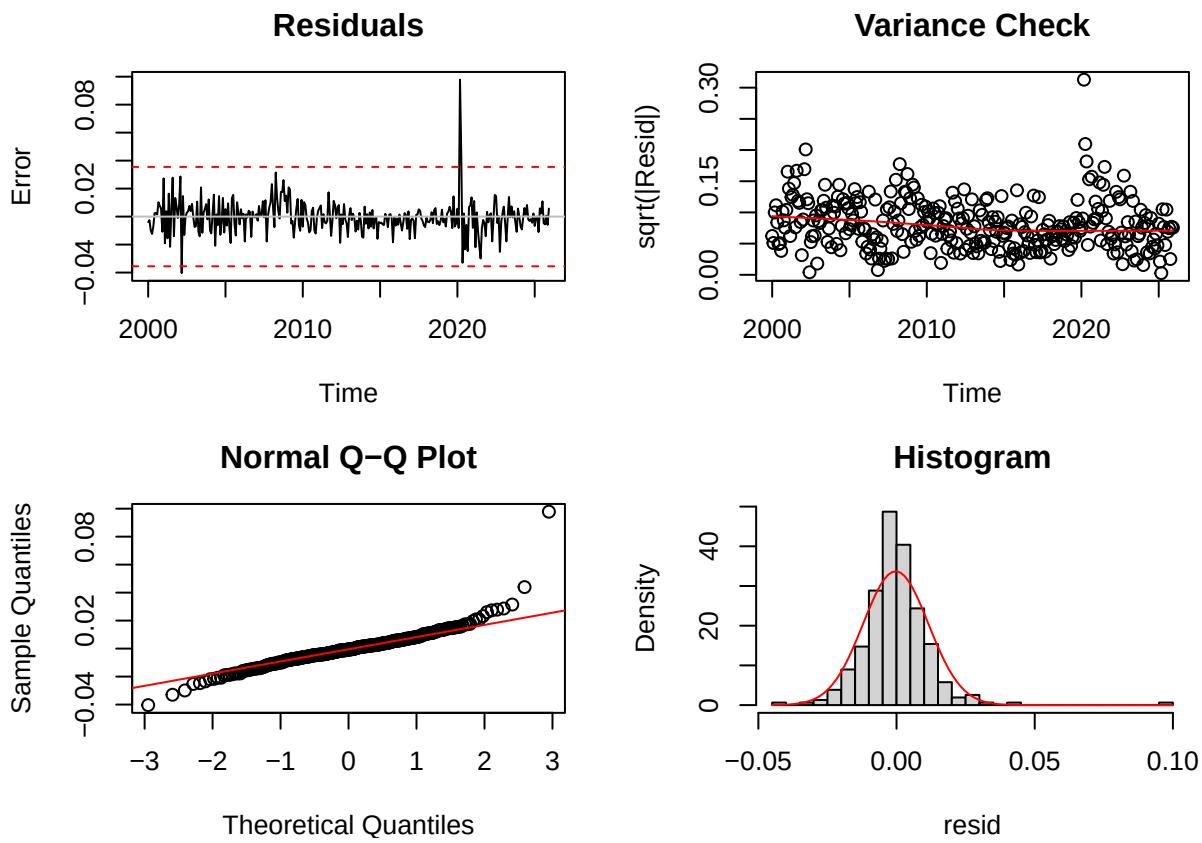

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(1,1,1)(2,0,2)12"
## AIC              "-1831.646"
## BIC              "-1805.468"
## Sigma2           "0.0001452655"
## Resid_Norm_PVal "1.013447e-14"
## Normal           "No"
## Resid_Ind_PVal   "0.3417723"
## Independent      "Yes"
## Min_AR_Root      "1.000709"
## Causal           "Yes"
## Min_MA_Root      "1.013097"
## Invertible       "Yes"
## RMSE             "0.0005420084"
## MAE              "0.0005062733"
## MAPE_Pct         "2.065196"
## Mean_CI_Width    "0.006162487"
##
## Processing Model 2 of 4...
##
## ====================================================================
##   ANALYSIS FOR MODEL: ARIMA(1,1,2)(2,0,2)12
## ====================================================================
```

```
##
##   CRITICAL WARNING: The Hessian matrix is unstable (Negative Variances).

## Warning in sqrt(diag_var): NaNs produced

##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##      Estimate Std.Error  t_stat p_value
## ar1    0.8250    0.1038  7.9489  0.0000
## ma1   -0.3144    0.1406 -2.2365  0.0253
## ma2   -0.0797    0.1048 -0.7612  0.4466
## sar1   0.2388        NA      NA      NA
## sar2   0.7469        NA      NA      NA
## sma1  -0.0963        NA      NA      NA
## sma2  -0.6472        NA      NA      NA
##
## Significance Flags:
##      Significant
## ar1          Yes
## ma1          Yes
## ma2           No
## sar1          No
## sar2          No
## sma1          No
## sma2          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000685
## Min MA Root Modulus:  1.013229
```



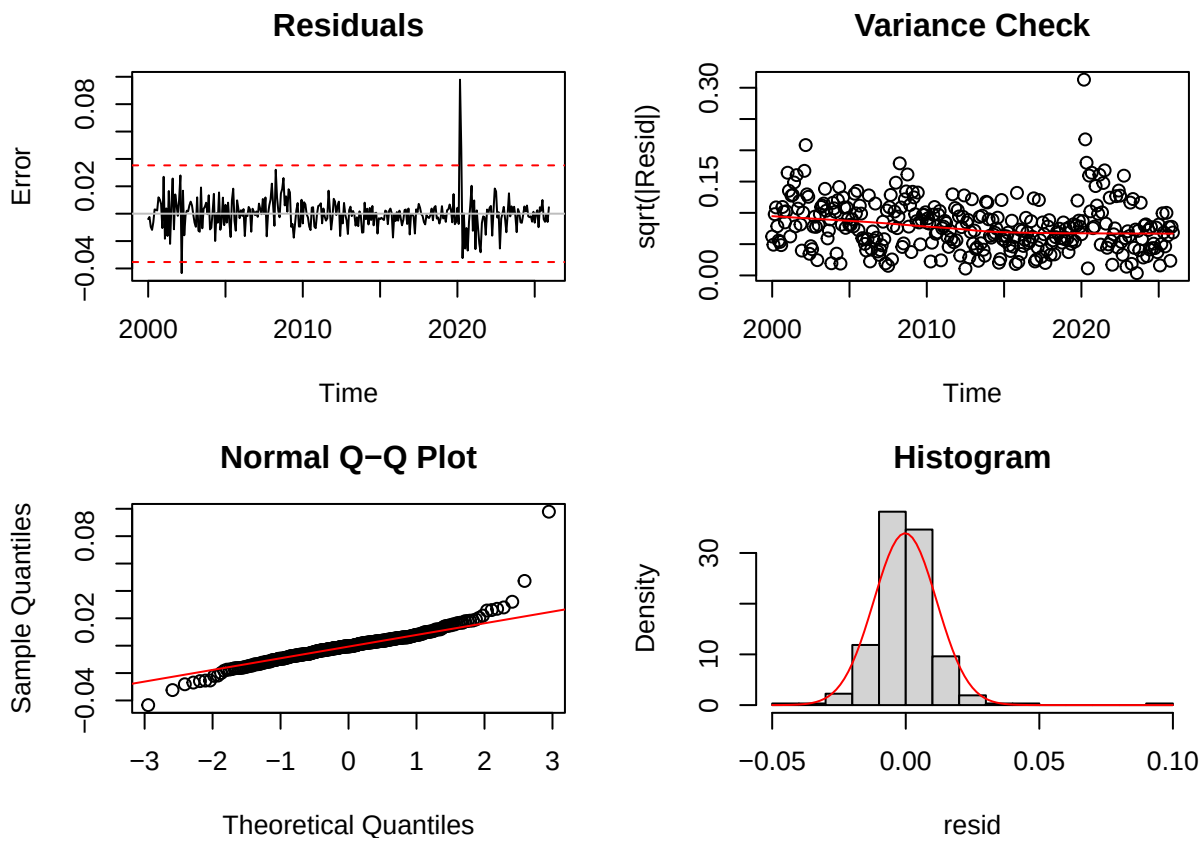

```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(1,1,2)(2,0,2)12"
## AIC              "-1830.092"
## BIC              "-1800.173"
## Sigma2           "0.0001448182"
## Resid_Norm_PVal "8.573002e-15"
## Normal           "No"
## Resid_Ind_PVal   "0.340921"
## Independent      "Yes"
## Min_AR_Root      "1.000685"
## Causal           "Yes"
## Min_MA_Root      "1.013229"
## Invertible       "Yes"
## RMSE             "0.0005113711"
## MAE              "0.0004783639"
## MAPE_Pct         "1.95021"
## Mean_CI_Width    "0.006101351"
##
## Processing Model 3 of 4...
##
## ================================================================
##   ANALYSIS FOR MODEL: ARIMA(0,1,9)(2,0,2)12
## ================================================================
##
##    CRITICAL WARNING: The Hessian matrix is unstable (Negative Variances).
```

```
## Warning in sqrt(diag_var): NaNs produced
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
##      Estimate Std.Error t_stat p_value
## ma1    0.5437    0.0577 9.4301  0.0000
## ma2    0.4294    0.0664 6.4629  0.0000
## ma3    0.2024    0.0703 2.8797  0.0040
## ma4    0.1271    0.0726 1.7512  0.0799
## ma5    0.1745    0.0690 2.5278  0.0115
## ma6    0.1989    0.0610 3.2597  0.0011
## ma7    0.0339    0.0601 0.5643  0.5726
## ma8    0.0787    0.0651 1.2090  0.2267
## ma9    0.0431    0.0640 0.6732  0.5008
## sar1   0.1896        NA     NA      NA
## sar2   0.7967        NA     NA      NA
## sma1  -0.0552        NA     NA      NA
## sma2  -0.6855        NA     NA      NA
##
## Significance Flags:
##      Significant
## ma1          Yes
## ma2          Yes
## ma3          Yes
## ma4           No
## ma5          Yes
## ma6          Yes
## ma7           No
## ma8           No
## ma9           No
## sar1          No
## sar2          No
## sma1          No
## sma2          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.00064
## Min MA Root Modulus:  1.01304
```




```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(0,1,9)(2,0,2)12"
## AIC              "-1827.696"
## BIC              "-1775.339"
## Sigma2           "0.0001400867"
## Resid_Norm_PVal "1.212919e-14"
## Normal           "No"
## Resid_Ind_PVal   "0.8486296"
## Independent      "Yes"
## Min_AR_Root      "1.00064"
## Causal           "Yes"
## Min_MA_Root      "1.01304"
## Invertible       "Yes"
## RMSE             "0.000552461"
## MAE              "0.0005213771"
## MAPE_Pct         "2.124051"
## Mean_CI_Width    "0.005922243"
##
## Processing Model 4 of 4...
##
## ==================================================================
##   ANALYSIS FOR MODEL: ARIMA(7,1,0)(2,0,2)12
## ==================================================================
##
## [1] COEFFICIENT SIGNIFICANCE (Alpha = 0.05 or |t| > 2):
```

```
##      Estimate Std.Error  t_stat p_value
## ar1    0.5155    0.0575  8.9678  0.0000
## ar2    0.1408    0.0637  2.2113  0.0270
## ar3   -0.0794    0.0637 -1.2469  0.2124
## ar4    0.0190    0.0646  0.2950  0.7680
## ar5    0.1273    0.0638  1.9963  0.0459
## ar6    0.0605    0.0636  0.9505  0.3418
## ar7   -0.0657    0.0570 -1.1527  0.2490
## sar1   1.3522    0.4988  2.7109  0.0067
## sar2  -0.3549    0.4955 -0.7163  0.4738
## sma1  -1.1706    0.5115 -2.2887  0.0221
## sma2   0.2397    0.4434  0.5405  0.5888
##
## Significance Flags:
##      Significant
## ar1          Yes
## ar2          Yes
## ar3           No
## ar4           No
## ar5          Yes
## ar6           No
## ar7           No
## sar1         Yes
## sar2          No
## sma1         Yes
## sma2          No
##
## [2] DIAGNOSTICS:
## Min AR Root Modulus:  1.000358
## Min MA Root Modulus:  1.008251
```




```
##
## [3] PERFORMANCE SUMMARY:
##                  [,1]
## Model_Name       "ARIMA(7,1,0)(2,0,2)12"
## AIC              "-1833.354"
## BIC              "-1788.476"
## Sigma2           "0.0001385058"
## Resid_Norm_PVal "3.003146e-15"
## Normal           "No"
## Resid_Ind_PVal   "0.7963259"
## Independent      "Yes"
## Min_AR_Root      "1.000358"
## Causal           "Yes"
## Min_MA_Root      "1.008251"
## Invertible       "Yes"
## RMSE             "0.0005068451"
## MAE              "0.0004770757"
## MAPE_Pct         "1.94307"
## Mean_CI_Width    "0.006014848"
##
## --- Comparison Complete ---
```

```r
print(comparison_table)
```

```
##               Model_Name       AIC       BIC       Sigma2 Resid_Norm_PVal Normal
## 1 ARIMA(1,1,1)(2,0,2)12 -1831.646 -1805.468 0.0001452655    1.013447e-14     No
## 2 ARIMA(1,1,2)(2,0,2)12 -1830.092 -1800.173 0.0001448182    8.573002e-15     No
## 4 ARIMA(7,1,0)(2,0,2)12 -1833.354 -1788.476 0.0001385058    3.003146e-15     No
```

```
## 3 ARIMA(0,1,9)(2,0,2)12 -1827.696 -1775.339 0.0001400867    1.212919e-14     No
##   Resid_Ind_PVal Independent Min_AR_Root Causal Min_MA_Root Invertible
## 1      0.3417723         Yes    1.000709    Yes    1.013097        Yes
## 2      0.3409210         Yes    1.000685    Yes    1.013229        Yes
## 4      0.7963259         Yes    1.000358    Yes    1.008251        Yes
## 3      0.8486296         Yes    1.000640    Yes    1.013040        Yes
##           RMSE          MAE MAPE_Pct Mean_CI_Width
## 1 0.0005420084 0.0005062733 2.065196   0.006162487
## 2 0.0005113711 0.0004783639 1.950210   0.006101351
## 4 0.0005068451 0.0004770757 1.943070   0.006014848
## 3 0.0005524610 0.0005213771 2.124051   0.005922243
```